Strategy research (research period 2012-2021 only)

1. The grid I tried (every variant, nothing hidden)

k	hold	side	gross Sharpe	net Sharpe	Sharpe 12-16	Sharpe 17-21	CAGR %	MaxDD %	trades	time in market %
0.5	1	both	1.62	0.97	1.16	0.82	13.05	-14.95	860	54.91
0.5	1	long	1.47	0.96	1.19	0.78	10.58	-18.72	563	39.73
0.5	2	both	1.68	1.25	1.21	1.28	20.19	-16.74	661	77.7
0.5	2	long	1.47	1.17	1.19	1.16	15.76	-19.67	385	57.16
0.5	3	both	1.44	1.09	1.11	1.07	18.13	-23.32	578	88.13
0.5	3	long	1.23	0.99	0.97	1.02	13.85	-20.18	318	64.98
1	1	both	1.58	1.07	1.21	0.98	10.93	-15.48	511	26.35
1	1	long	1.52	1.08	1.03	1.13	8.48	-8.8	340	18.29
1	2	both	1.47	1.13	1.01	1.24	14.82	-9.55	430	43.78
1	2	long	1.44	1.16	1.1	1.22	11.71	-8.93	273	30.69
1	3	both	1.17	0.9	0.94	0.87	12.53	-14.07	381	55.77
1	3	long	1.2	0.99	1.21	0.83	11.04	-10.43	233	39.4
1.5	1	both	1.01	0.66	0.74	0.62	4.93	-13.71	269	12.36
1.5	1	long	1.04	0.76	0.57	0.91	4.43	-6.78	167	7.82
1.5	2	both	1.23	1	0.98	1.03	10.5	-12.91	244	21.89
1.5	2	long	1.22	1.03	1.19	0.9	7.63	-9.72	147	14.12
1.5	3	both	1.07	0.88	1.08	0.75	9.92	-16.08	223	29.87
1.5	3	long	1.06	0.9	1.16	0.72	7.52	-11.39	132	19.31

Variants passing the rule (>= 100 trades, positive net Sharpe in both halves): 18 of 18
Chosen: k=0.5, hold=2 day(s), side=both, window=60

2. Did I just pick the luckiest of 18 variants? (White's Reality Check)

Bootstrap p-value that the best of the 18 variants is no better than zero-edge luck: **0.003**
(small = the best variant is hard to explain by selection luck alone)

3. Chosen rule, research period

	Strategy (net of costs)	Strategy 2012-16	Strategy 2017-21	Buy & hold Nifty (open to open)
CAGR %	20.19	17.89	22.5	14.4
Ann vol %	15.76	14.43	16.98	17.64
Sharpe	1.25	1.21	1.28	0.85
Max drawdown %	-16.74	-16.74	-11.57	-37.77
Days	2444	1216	1228	2444
Trades	661	333	328	nan
Win rate %	51.89	51.05	52.74	nan
Avg win %	1.55	1.53	1.57	nan
Avg loss %	-1.07	-1.08	-1.07	nan

Gross Sharpe (before costs): 1.68. Total costs paid over the period: 66.1% of capital (simple sum of daily costs).
Time in the market: 78% of days. Average trade: 0.29% net.

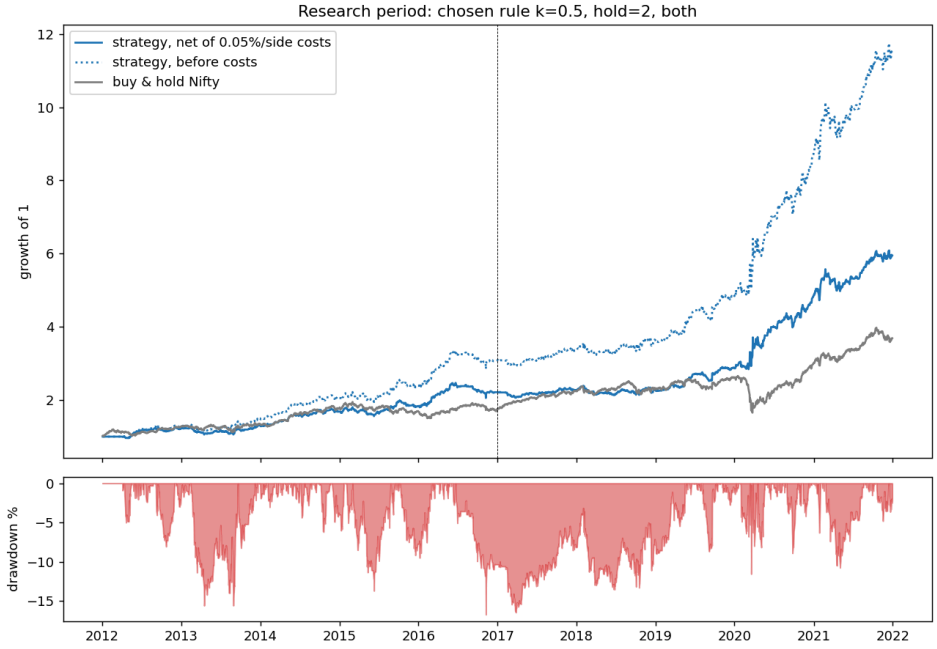
Net return by calendar year (%):

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
net return %	22.5	6.6	28.9	8.9	20.7	2.4	0.7	25.7	68.5	23.1

Gross P&L split: long side 189.0% of capital, short side 67.5% of capital (sum of daily gross, no compounding).
Net Sharpe excluding 2020: 1.13 (full period 1.25)

Same rule, same parameters on **Bank Nifty** (no re-tuning): net Sharpe 0.79, CAGR 16.7%, max DD -26.7%, 687 trades.

The chart shows the growth of 1 for the chosen rule after costs (solid) and before costs (dotted) against buy and hold, with the drawdown of the rule underneath. The thin vertical line at the start of 2017 separates the two halves used for the half-period Sharpe columns above.



4. How much do costs matter?

cost per side %	net Sharpe	CAGR %
0	1.68	28.65
0.02	1.46	24.35
0.05	1.25	20.19
0.08	1.03	16.16
0.1	0.81	12.27

5. Parameter sensitivity (each parameter moved by -20% / +20%)

Hold days is an integer so I moved it by one day (h-1, h+1); with a 1-3 day hold, 20% is less than a day. The window is rounded to whole days.

variant	k	window	hold	net Sharpe	CAGR %	MaxDD %	trades
base	0.5	60	2	1.25	20.19	-16.74	661
k -20%	0.4	60	2	0.99	16.01	-19.38	716
k +20%	0.6	60	2	1.23	19.34	-18.6	615
window -20%	0.5	48	2	1.15	18.56	-17.27	672
window +20%	0.5	72	2	1.32	21.67	-16.27	654
hold -1	0.5	60	1	0.97	13.05	-14.95	860
hold +1	0.5	60	3	1.09	18.13	-23.32	578
corner k-20% w-20% h-1	0.4	48	1	0.73	10.04	-16.47	936
corner k-20% w-20% h+1	0.4	48	3	0.91	15.16	-25.34	653
corner k-20% w+20% h-1	0.4	72	1	0.87	12.17	-13.65	909
corner k-20% w+20% h+1	0.4	72	3	1	16.62	-25.24	637
corner k+20% w-20% h-1	0.6	48	1	0.95	12.34	-14.47	787
corner k+20% w-20% h+1	0.6	48	3	1.05	17.23	-22.98	537
corner k+20% w+20% h-1	0.6	72	1	0.96	12.41	-15.48	755
corner k+20% w+20% h+1	0.6	72	3	1.01	16.26	-26.42	521

Net Sharpe is positive in 100% of these 15 variants. Range: 0.73 to 1.32.

6. Drawdown I should realistically expect (stationary bootstrap, 10,000 paths)

Over a 1000-day (~4 year) horizon, resampling this strategy's own daily net returns.

Scenario 1: the edge is exactly what I measured in the research period

	max drawdown %	annualised return %	Sharpe
typical (median)	-14.9	20	1.3
bad (worst 25%)	-18.1	14.4	1
worse (worst 10%)	-21.9	9.5	0.7
very bad (worst 5%)	-24.6	6.7	0.5
extreme (worst 1%)	-29.6	1.9	0.2

P(4-year max drawdown worse than 10%) = 96%, worse than 15% = 50%, worse than 20% = 16%, worse than 25% = 4%. P(4-year Sharpe <= 0) = 0.2%.

Scenario 2: only half of the measured edge is real (daily mean return cut by 50%)

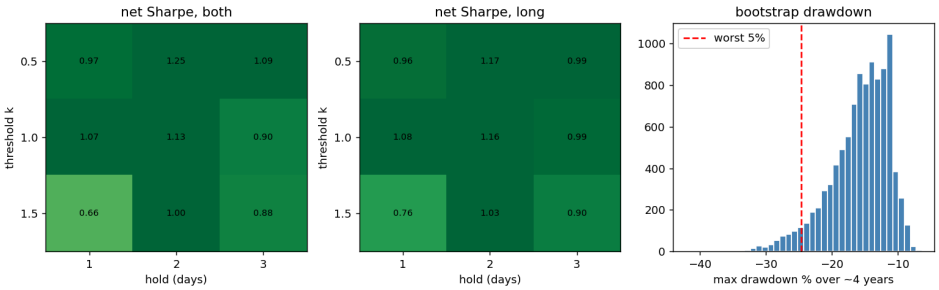
	max drawdown %	annualised return %	Sharpe
typical (median)	-19.8	8.8	0.6
bad (worst 25%)	-24.8	3.7	0.3
worse (worst 10%)	-30.6	-0.7	0
very bad (worst 5%)	-34.2	-3.3	-0.2
extreme (worst 1%)	-40.9	-7.6	-0.5

P(4-year max drawdown worse than 10%) = 99%, worse than 15% = 81%, worse than 20% = 49%, worse than 25% = 24%. P(4-year Sharpe <= 0) = 9.2%.

Historical max drawdown of the rule in the research period: -16.7%.

Caveat: the bootstrap only captures sampling noise. It cannot see regime change or the selection bias of picking this rule out of many ideas, so scenario 2 is the one i would plan around.

Left and middle: net Sharpe of the 18 variants of the grid, trading both sides and long only (threshold k down, holding days across). Right: the bootstrap distribution of the worst drawdown over about four years, with the worst 5% of paths marked.



7. How much is just market exposure?

Average net exposure 0.37 of capital. Beta to always-long Nifty 0.33 (t=4.6). Alpha 14.7% a year (t=3.02).